NVIDIA Unreal Engine DLSS plugins

8.8.0 September 2026

Component	Note
Unreal Engine	5.8 to 5.6, deprecated: 5.5
NVIDIA RTX	5.8
UE Platforms	Win64 (x86_64), WinGDK
Streamline SDK	2.14.1
NGX SDK	310.9.1

RHI Support:

- D3D12RHI, D3D11RHI, VulkanRHI
    - DLSS Super Resolution (DLSS-SR)
    - Deep Learning Anti Aliasing (DLAA)
    - Image Scaling (NIS)
- D3D12RHI, D3D11RHI
    - RTX Dynamic Vibrance
    - Reflex Low Latency
- D3D12RHI
    - DLSS Ray Reconstruction (DLSS-RR)
    - DLSS Frame Generation (DLSS-FG)
    - DLSS Multi Frame Generation (DLSS-MFG)

Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

Note: The NVIDIA Image Scaling plugin is RHI agnostic, but might need adjustments for private platforms like consoles since only public RHIs are tested

Note: Support for deprecated UE versions will be removed when a new UE 5.X version gets supported

New Features

- Added `DLSS Deferred Renderer` node (`UMovieGraphDLSSRenderPassNode`) for Movie Render Graph, enabling DLSS-SR and DLAA during offline rendering via the graph-based render pipeline introduced in UE 5.4
- Added DLSS-RR preset F.

General Changes

- Internal plugins StreamlineCore and StreamlineNGXCommon are now correctly hidden from the editor plugins menu. They were always meant to be hidden, as they should be automatically included in your project by adding the feature plugins they depend on such as StreamlineDLSSFG.

Breaking Changes

- UStreamlineLibrary::IsStreamlineFeatureSupported, UStreamlineLibrary::QueryStreamlineFeatureSupport, UStreamlineLibraryReflex::IsReflexSupported, UStreamlineLibraryReflex::QueryReflexSupport, and the associated blueprint functions were reporting Reflex as supported even on configurations that don't support Reflex including non-NVIDIA hardware. Reporting has been fixed to be accurate now. If any applications were relying on these functions reporting support for non-NVIDIA hardware, for example to determine if PCL (PC Latency) markers are available, the application logic will need to be modified.
- DLSS-SR/RR auto quality mode is explicitly not supported in Play-In-Editor (PIE) windows. It was previously reported as supported, but it's not generally possible to get an accurate viewport size from blueprints when running in PIE so applications would implement it improperly.

Reflex Low Latency

- UStreamlineLibrary::IsStreamlineFeatureSupported, UStreamlineLibrary::QueryStreamlineFeatureSupport, UStreamlineLibraryReflex::IsReflexSupported, UStreamlineLibraryReflex::QueryReflexSupport, and the associated blueprint functions were reporting Reflex as supported even on configurations that don't support Reflex including non-NVIDIA hardware. Reporting has been fixed

to be accurate now.

DLSS Super Resolution (DLSS-SR)

- Added `UMovieGraphDLSSRenderPassNode` (**DLSS Deferred Renderer** node) for Movie Render Graph. Place this node in a Movie Render Graph in place of the standard Deferred Renderer node to apply DLSS-SR or DLAA upscaling during offline rendering. The desired quality preset (DLAA, Ultra Quality, Quality, Balanced, Performance, Ultra Performance) is set via the node's Details panel. Requires the *Movie Render Queue DLSS Support* plugin to be enabled.
- Removed DLSS-SR/RR auto quality mode in Play-In-Editor (PIE) windows.

DLSS Ray Reconstruction (DLSS-RR)

- Added preset F (default).

DLSS Frame Generation (DLSS-FG)

- Fixed DLSS-FG when using `r.D3D12.PresentOnSubmissionThread` mode that was added in Unreal 5.8.
- Experimental fix for DLSS-FG when using dynamic resolution. Set the `r.Streamline.DLSSG.DynamicResolutionMode` console variable to `2` to automatically adapt DLSS-FG to the engine's current dynamic resolution state.

RTX Dynamic Vibrance

- Removed the "stat deepdvc" console command that didn't do anything useful.

NVIDIA Image Scaling

- Fixed shader compilation error with older GPUs that don't support native 16-bit float ops

Bug Fixes

- 6533447 - Fixed DLSS-FG when using `r.D3D12.PresentOnSubmissionThread` mode that was added in Unreal 5.8.
- UStreamlineLibrary::IsStreamlineFeatureSupported, UStreamlineLibrary::QueryStreamlineFeatureSupport, UStreamlineLibraryReflex::IsReflexSupported, UStreamlineLibraryReflex::QueryReflexSupport, and the associated blueprint functions were reporting Reflex as supported even on configurations that don't support Reflex including non-NVIDIA hardware. Reporting has been fixed to be accurate now.
- NIS: Fixed shader compilation error with older GPUs that don't support native 16-bit float ops
- Experimental fix for DLSS-FG when using dynamic resolution.
- 6405516 - Removed redundant "OFF" selections in DLSS4Sample app Reflex menu.
- 5870805 - [DX12][DLSS-FG][Intermittent] - Application crash observed shortly after launch (5.5, 5.6)
- 5417010 - [DX12][UE 5.6][Intermittent] - Stand alone game mobile crashes on launch

Known Issues

General:

- 6713943 - [UE 5.8] - Enabling DLSS-FG creates garbage reflex latency numbers
- 6706974 - [UE 5.8] - DLSS-FG doesn't run every frame
- "Use SetWithCurrentPriority or ReplaceCurrentPriorityAndTag with explicit MinPriority and MaxPriority values" log messages
- 6143189 - [UE 5.7/5.6/5.5] - -AttachRenderDoc does not populate warn message
- 6142757 - [UE 5.7/5.5] - GPU crash with Movie Render Queue and Streamline debug overlay enabled
- 5504512 - [DLSS4 Test App] - Heavily amplified instability observed with DLSS-SR Ultra Performance option (UE 5.6 & 5.5)
- 5416670 - [DLSS4 Test App][Intermittent] - Stat DLSS average num continuously fluctuates with DLSS-SR enabled (UE 5.5, 5.6)
- 5356855 - Running the packaged sample in vulkan mode causes significant render instability or an immediate crash

Editor Specific:

- 6699882 - [UE 5.8] - Editor crashes after enabling FG and resizing editor window
- 5076953 - Editor crashes with resolution/screen mode changes in Standalone Game mode with multi-display

DLSS4Sample App Specific:

- New: 6712305 - [UE DLSS 8.8.0/SL v.2.14.0][UE 5.8/5.7/5.6][DLSS4Test] - Reflex options are missing in DX11

8.7.2 July 2026

Hotfix: Restore missing DLSS-SR HDR flag, which corrects image quality issues with the 8.7.0 release.

8.7.0 July 2026

Component	Note
Unreal Engine	5.8 to 5.6 deprecated: 5.5
NVIDIA RTX	5.8
UE Platforms	Win64, WinGDK
Streamline SDK	2.11.1
NGX SDK	310.6.0

RHI Support:

- D3D12RHI, D3D11RHI, VulkanRHI
    - DLSS Super Resolution (DLSS-SR)
    - Deep Learning Anti Aliasing (DLAA)
    - Image Scaling (NIS)
- D3D12RHI, D3D11RHI
    - RTX Dynamic Vibrance
    - Reflex Low Latency
- D3D12RHI
    - DLSS Ray Reconstruction (DLSS-RR)
    - DLSS Frame Generation (DLSS-FG)
    - DLSS Multi Frame Generation (DLSS-MFG)

Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

Note: The NVIDIA Image Scaling plugin is RHI agnostic, but might need adjustments for private platforms like consoles since only public RHIs are tested

Note: Support for deprecated UE versions will be removed when a new UE 5.X version gets supported

New Features

- Restored DLSS-SR legacy preset E (note: release notes for 8.6 plugin claimed this was restored but the change was not actually made until 8.7)

General Changes

- Added Unreal 5.8 support.
- Renamed some config files from Base*.ini to Default*.ini to match current engine conventions.

Breaking Changes

Nothing new in this release

Reflex Low Latency

Nothing new in this release

DLSS Super Resolution (DLSS-SR)

- Restored legacy preset E, which had previously been removed in version 8.3.0 plugin

DLSS Ray Reconstruction (DLSS-RR)

- GBuffer resolve pass was changed from a pixel shader to a compute shader. There is no expected difference in behavior, but if any issues occur the old pixel shader path can be restored by setting the r.NGX.DLSS.GBufferResolve.Compute cvar to false.

DLSS Frame Generation (DLSS-FG)

Nothing new in this release

RTX Dynamic Vibrance

Nothing new in this release

NVIDIA Image Scaling

Nothing new in this release

Bug Fixes

- Fix missing binaries when using WinGDK platform extension
- Fix a nuisance assertion for some resolutions when DLSS-SR was enabled. The assertion was a test for the screen percentage being within an acceptable range for DLSS-SR, but due to quantization some ViewRects could occasionally fail the test. This release increases the tolerance for the check and changes it to a non-fatal ensure.
- Fix error when compiling with clang on Windows

Known Issues

General:

- New - 6713943 - [UE 5.8] - Enabling DLSS-FG creates garbage reflex latency numbers
- New - 6706974 - [UE 5.8] - DLSS-FG doesn't run every frame
- 6143189 - [UE 5.7/5.6/5.5] - -AttachRenderDoc does not populate warn message
- 6142757 - [UE 5.7/5.5] - GPU crash with Movie Render Queue and Streamline debug overlay enabled
- 5870805 - [DX12][DLSS-FG][Intermittent] - Application crash observed shortly after launch (5.5, 5.6)
- 5504512 - [DLSS4 Test App] - Heavily amplified instability observed with DLSS-SR Ultra Performance option (UE 5.6 & 5.5)
- 5417010 - [DX12][UE 5.6][Intermittent] - Stand alone game mobile crashes on launch
- 5416670 - [DLSS4 Test App][Intermittent] - Stat DLSS average num continuously fluctuates with DLSS-SR enabled (UE 5.5, 5.6)
- 5356855 - Running the packaged sample in vulkan mode causes significant render instability or an immediate crash

Editor Specific:

- New: 6699882 - [UE 5.8] - Editor crashes after enabling FG and resizing editor window
- 5076953 - Editor crashes with resolution/screen mode changes in Standalone Game mode with multi-display
- 4202194 - Unchecking "Warn about incompatible plugin tools" on VK does not remove the dialog box, even after restarting the editor

8.6.1 May 2026

Hotfix: DLSS-FG For vsync support please use GetDLSSGIsVsyncSupportAvailable function instead of the result of calling GetStreamlineFeatureInformation.

DLSS Frame Generation (DLSS-FG)

- Added GetDLSSGIsVsyncSupportAvailable BP function.

8.6.0 May 2026

Component	Note
Unreal Engine	5.7 to 5.5 deprecated: 5.4
NVIDIA RTX	5.7
UE Platforms	Win64, WinGDK
Streamline SDK	2.11.1
NGX SDK	310.6.0

RHI Support:

- D3D12RHI, D3D11RHI, VulkanRHI
    - DLSS Super Resolution (DLSS-SR)
    - Deep Learning Anti Aliasing (DLAA)
    - Image Scaling (NIS)
- D3D12RHI, D3D11RHI
    - RTX Dynamic Vibrance

    - Reflex Low Latency

- D3D12RHI

    - DLSS Ray Reconstruction (DLSS-RR)
    - DLSS Frame Generation (DLSS-FG)
    - DLSS Multi Frame Generation (DLSS-MFG)

Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

Note: The NVIDIA Image Scaling plugin is RHI agnostic, but might need adjustments for private platforms like consoles since only public RHIs are tested

Note: Support for deprecated UE versions will be removed when a new UE 5.X version gets supported

New Features

- Updated to SL SDK 2.11.1
- Updated to NGX SDK 310.6.0
- Added DLSS-MFG 5x and 6x modes
- Added DLSS-MFG dynamic mode
- Added DLSS-FG/MFG support for Vsync
- Added DLSS-SR exposure scale

General Changes

- Use UE plugin side DXGI swapchain proxy and custom present for swapchain tracking for DLSS-FG instead of Slate callbacks
    - On UE 5.8+ the slate callbacks are removed from the engine and the swapchain proxy is always used
    - On UE 5.7 and older the Slate callback path is still available but deprecated
        - Fallback to old behavior can be used with `Use Slate callbacks for Swapchain tracking (deprecated)` plugin and plugin override setting as well as -sl{no}slatecallbacks command line arguments
    - Moved `FStreamlineViewExtension` view tracking methods from StreamlineViewExtension.cpp into separate StreamlineViewTracking.cpp file
    - This is incompatible with other features that also use a custom present.
- Remove some nuisance Streamline log messages about unsupported features

Breaking Changes

- DLSS-SR and DLSS-RR disabled by default to improve compatibility with 3rd party upscaler plugins
    - Existing BP logic should explicitly call `UDLSSLibrary::{EnableDLSS, EnableDLAA, EnableDLSSRR}`
- Restrict DLSS-FG to only a single "New Editor Window (PIE)" window

Reflex Low Latency

- Fix possible crash during shutdown if another latency marker provider or max tick rate handler is registered.

DLSS Super Resolution (DLSS-SR)

- r.NGX.DLSS.Enable default changed to 0
- Added support for DLSS-SR exposure scale setting through r.NGX.DLSS.ExposureScale cvar
- Set r.TemporalAA.Upsampling to 1 when DLSS-SR is enabled, in case another feature has disabled it

DLSS Ray Reconstruction (DLSS-RR)

Nothing new in this release

DLSS Frame Generation (DLSS-FG)

- Disabled DLSS-FG/MFG for when multiple PIE New Editor windows are open, i.e. when PlayNumberOfClients is not 1
- Added DLSS-MFG 5x and 6x modes. Only available on supported hardware, generally RTX 50 series or higher.
- Added DLSS-MFG dynamic mode, which dynamically enables multi frame generation. This replaces DLSS-FG auto mode on newer hardware, generally RTX 50 series or higher.
- Added DLSS-FG/MFG support for Vsync in some cases. Use GetStreamlineFeatureInformation and BreakStreamlineFeatureRequirements from blueprints or from C++ to query at runtime whether Vsync is supported.

RTX Dynamic Vibrance

Nothing new in this release

NVIDIA Image Scaling

Nothing new in this release

Bug Fixes

- Fix possible crash during shutdown if another latency marker provider or max tick rate handler is registered.
- 5871568 - Use project override option is not functional while warn about incompatible plugins and tools is disabled
- 6043733 - DLSS-SR will not engage after switching to NIS and back

Known Issues

General:

- New: 6143189 - [UE 5.7/5.6/5.5] - -AttachRenderDoc does not populate warn message
- New: 6142757 - [UE 5.7/5.5] - GPU crash with Movie Render Queue and Streamline debug overlay enabled
- 5870805 - [DX12][DLSS-FG][Intermittent] - Application crash observed shortly after launch (5.5, 5.6)
- 5504512 - [DLSS4 Test App] - Heavily amplified instability observed with DLSS-SR Ultra Performance option (UE 5.6 & 5.5)
- 5417010 - [DX12][UE 5.6][Intermittent] - Stand alone game mobile crashes on launch
- 5416670 - [DLSS4 Test App][Intermittent] - Stat DLSS average num continuously fluctuates with DLSS-SR enabled (UE 5.5, 5.6)
- 5356855 - Running the packaged sample in vulkan mode causes significant render instability or an immediate crash

Editor Specific:

- 5076953 - Editor crashes with resolution/screen mode changes in Standalone Game mode with multi-display
- 4202194 - Unchecking "Warn about incompatible plugin tools" on VK does not remove the dialog box, even after restarting the editor

8.5.0 February 2026

Component	Note
Unreal Engine	5.7 to 5.5 deprecated: 5.4
NVIDIA RTX	5.7
UE Platforms	Win64, WinGDK
Streamline SDK	2.10.3
NGX SDK	310.5.0

RHI Support:

- D3D12RHI, D3D11RHI, VulkanRHI
    - DLSS Super Resolution (DLSS-SR)
    - Deep Learning Anti Aliasing (DLAA)
    - Image Scaling (NIS)
- D3D12RHI, D3D11RHI
    - RTX Dynamic Vibrance
    - Reflex Low Latency
- D3D12RHI
    - DLSS Ray Reconstruction (DLSS-RR)
    - DLSS Frame Generation (DLSS-FG)
    - DLSS Multi Frame Generation (DLSS-MFG)

Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

Note: The NVIDIA Image Scaling plugin is RHI agnostic, but might need adjustments for private platforms like consoles since only public RHIs are tested

Note: Support for deprecated UE versions will be removed when a new UE 5.X version gets supported

New Features

- Updated to SL SDK 2.10.3
- Updated to NGX SDK 310.5.0

General Changes

Breaking Changes

- DLSS-SR: removed dilated, hi-resolution motion vectors support to match SDK recommendations for recent transformer based DLSS-SR models.
    - This brings DLSS-SR in line with DLSS-RR.
    - This is expected to improve image quality.
- added DLSS-SR presets L and M, deprecated preset F

Reflex Low Latency

Nothing new in this release

DLSS Super Resolution (DLSS-SR)

- removed dilated hi resolution motion vectors.
    - removed r.NGX.DLSS.DilateMotionVectors cvar and related code
    - current and recent DLSS-SR presets are optimized for low resolution motion vectors
- deprecated DLSS-SR preset F
- added DLSS-SR presets L and M

DLSS Ray Reconstruction (DLSS-RR)

Nothing new in this release

DLSS Frame Generation (DLSS-FG)

Nothing new in this release

RTX Dynamic Vibrance

Nothing new in this release

NVIDIA Image Scaling

Nothing new in this release

Bug Fixes

- 5718115 - Intermittent UE Editor Crashes observed with Vulkan API (5.5, 5.7)
- 5717157 - Intermittent Unreal Engine Editor crashes upon changing screen percentage with all viewports launched (5.6)
- 5718096 - [DLSS4Test] - Missing DLSS settings in Selected Viewport, PIE, and Standalone modes (5.5)
- 5718081 - PIE mode crashes on launch with "Load Debug Overlay" option selected (5.5)
- 5717857 - [DX12][UE 5.6/5.7] - Unreal Engine DLSS4Test application crashes on launch with Non-NV GPUs

Known Issues

General:

- New: 5870805 - [DX12][DLSS-FG][Intermittent] - Application crash observed shortly after launch (5.5, 5.6)
- New: 5817157 - [DX11/DX12] UE crash error with: DLSS4Test.exe -dx11 and/or -dx12 -d3ddebug -rhivalidation checking (5.4.4)
- New: 5814005 - [Vulkan] - UE Editor crashes after applying perspective/lit & real time in 2x2 layout (5.6)
- 5504512 - [DLSS4 Test App] - Heavily amplified instability observed with DLSS-SR Ultra Performance option (UE 5.6 & 5.5)
- 5417010 - [DX12][UE 5.6][Intermittent] - Stand-alone game crashes on launch
- 5416670 - [DLSS4 Test App][Intermittent] - Stat DLSS average num continuously fluctuates with DLSS-SR enabled (UE 5.5, 5.6)
- 5356855 - Running the packaged sample in vulkan mode causes significant render instability or an immediate crash

Editor Specific:

- 5417003 - Editor crashes after MovieRenderQueue ends (UE 5.6)
- 5076953 - Editor crashes with resolution/screen mode changes in Standalone Game mode with multi-display
- 4202194 - Unchecking "Warn about incompatible plugin tools" on VK does not remove the dialog box, even after restarting the editor

DLSS4Test App Specific:

- New: 5816879 - [DX11/Vulkan] Unable to toggle different resolutions (only 1080p works) (5.4.4)
- New: 5817138 - Mouse cursor does not match resolution with windowed mode + 1080p/1440p (5.4.4)
- 5718096 - [DX12][DX11][Vulkan][UE 5.5] - Missing DLSS settings in Selected Viewport, PIE, and Standalone modes
- 5718060 - [DX12][DX11][UE 5.7/5.5] - Launching DLSS4Test.exe with -sldebugoverlay causes SL to not function and the overlay does not appear
- 5426798 - [DLSS-RR] - Severe exposure corruption observed with different View Rect Offset options while DLSS-RR is enabled
- 5502683 - [Vulkan] - DLSS4Test-Win64-Shipping crashes on launch with non-NV GPUs (UE 5.6)
- 5502626 - [Vulkan] - DLSS4Test application exhibits heavy corruption on non-NV GPUs (UE 5.6)
- 5502713 - Application crash observed upon cycling through 'constrained view' and 'View Rect Offset' options with DLSS-SR performance modes (UE 5.6)
- 5417032 - [DX12] - Application launches to blank window on non-NV GPUs (UE 5.5)
- 5228046 - [RTX] - Image corruption occurs when View Rect Offset is adjusted
- 5416464 - Application exhibits heavy corruption
- 5416646 - DLSS-SR is not engaged on launch despite 'auto' mode being selected
- 5416702 - [DX12][DLSS-FG] - Heavy stutter observed when changing constrained views with DLSS-FG enabled (UE 5.5)
- 5416990 - Only the DLSS4 Log populates when logging
- 5046562 - [DLSS-SR] - Reflection artifacts observed with DLSS-SR enabled in triple view
- 5046557 - Amplified instability observed on thin lines with different constrained views
- 5046507 - [DLSS-SR] - Amplified noise on reflections with DLSS-SR Enabled
- 5050485 - DLSS-SR + DLSS-RR UIs are functional with non-NV GPUs

8.4.0 December 2025

Component	Note
Unreal Engine	5.7 to 5.5 deprecated: 5.4
NVIDIA RTX	5.6
UE Platforms	Win64, WinGDK
Streamline SDK	2.9.0
NGX SDK	310.4.0

RHI Support:

- D3D12RHI, D3D11RHI, VulkanRHI
    - DLSS Super Resolution (DLSS-SR)
    - Deep Learning Anti Aliasing (DLAA)
    - Image Scaling (NIS)
- D3D12RHI, D3D11RHI
    - RTX Dynamic Vibrance
    - Reflex Low Latency
- D3D12RHI
    - DLSS Ray Reconstruction (DLSS-RR)
    - DLSS Frame Generation (DLSS-FG)
    - DLSS Multi Frame Generation (DLSS-MFG)

Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

Note: The NVIDIA Image Scaling plugin is RHI agnostic, but might need adjustments for private platforms like consoles since only public RHIs are tested

Note: Support for deprecated UE versions will be removed when a new UE 5.X version gets supported

New Features

- Added support for UE 5.7
- Deprecated support for UE 5.4

General Changes

- Minor changes to support Engine changes in 5.7

- Added StreamlineNGXCommon plugin that contains code that is in common between multiple plugins to avoid code duplications.
- Replaced *DLSS4Test* sample in favor of a simpler *DLSS4Sample*
- moved `*_API` module exports from the whole class/type into the relevant methods
    - added `UE_API` macros scattered through most `.h` files

Breaking Changes

- Disabled SL & NGX plugins loading when running a non-cooker commandlet.
    - Added *Incompatibilities with Non-Cooker Commandlets* sections to the the plugin documentations
- refine logic to determine whether Reflex Low Latency is supported
    - this addresses a corner case where Reflex Low Latency was unavailable from the SL SDK perspective, but that was not propagated up to the application/UI via the UE Streamline Reflex plugin blueprint library
- Implement SL SDK guidelines for SL Reflex related to `slReflexSleep` and Reflex Low Latency mode being off
    - This improves compatibility with driver based FPS limiting settings

Reflex Low Latency

- refine logic to determine whether Reflex Low Latency is supported
    - In `FStreamlineLatencyBase::IsStreamlineReflexSupported()` check whether `slReflexGetState::lowLatencyAvailable` is true
    - This is in addition to the existing check of `slisFeatureSupported(sl::kFeatureReflex)`
- Implement SL SDK guidelines for SL Reflex related to `slReflexSleep` and Reflex Low Latency mode being off
    - Assumptions:
        - Reflex Low Latency is supported and the feature overall enabled via `t.Streamline.Reflex.Enable 1` or `t.Streamline.Reflex.EnableInEditor 1`
        - Reflex Low Latency *mode* is set *off*
            - explicitly via `t.Streamline.Reflex.Mode 0`
            - implicitly via `t.Streamline.Reflex.Auto` and DLSS-FG being off
    - New Behavior in 8.4.0
        - `slReflexSleep` is called
    - Previous Behavior, 8.3.0 and older
        - `slReflexSleep` is NOT called
    - Unchanged behavior
        - `slReflexSleep` is called when Reflex Low Latency *mode* is *on* via `t.Streamline.Reflex.Mode {1,2}` or `t.Streamline.Reflex.Auto`

DLSS Super Resolution (DLSS-SR)

Nothing new in this release

DLSS Ray Reconstruction (DLSS-RR)

Nothing new in this release

DLSS Frame Generation (DLSS-FG)

Nothing new in this release

RTX Dynamic Vibrance

Nothing new in this release

NVIDIA Image Scaling

Nothing new in this release

Bug Fixes

- UE-1496 World Partition HLOD Generation Fails with StreamlineCore Plugin in UE5.6

Known Issues

General:

- 5504512 - [DLSS4 Test App] - Heavily amplified instability observed with DLSS-SR Ultra Performance option (UE 5.6 & 5.5)
- 5417010 - [DX12][UE 5.6][Intermittent] - Stand-alone game crashes on launch

- 5416670 - [DLSS4 Test App][Intermittent] - Stat DLSS average num continuously fluctuates with DLSS-SR enabled (UE 5.5, 5.6)
- 5356855 - Running the packaged sample in vulkan mode causes significant render instability or an immediate crash

Editor Specific:

- New: 5718115 - [Vulkan][UE 5.7/5.5][Intermittent] - Intermittent UE Editor Crashes observed with Vulkan API
- New: 5717157 - [Vulkan][UE 5.6][Intermittent] - Unreal Engine Editor crashes upon changing screen percentage with all viewports launched
- New: 5718081 - [DX12][UE 5.5] - PIE mode crashes on launch with "Load Debug Overlay" option selected
- 5416522 - Unreal Engine Editor crashes when launched with Vulkan API (UE 5.2)
- 5417003 - Editor crashes after MovieRenderQueue ends (UE 5.6)
- 5417021 - Editor crashes after toggling through different DLSS-RR presets while playing the project in a new window (UE 5.2)
- 5076953 - Editor crashes with resolution/screen mode changes in Standalone Game mode with multi-display
- 4202194 - Unchecking "Warn about incompatible plugin tools" on VK does not remove the dialog box, even after restarting the editor

DLSS4Test App Specific:

- New: 5717857 - [DX12][UE 5.6/5.7] - Unreal Engine DLSS4Test application crashes on launch with Non-NV GPUs (Not a regression)
- New: 5718096 - [DX12][DX11][Vulkan][UE 5.5] - Missing DLSS settings in Selected Viewport, PIE, and Standalone modes
- New: 5718060 - [DX12][DX11][UE 5.7/5.5] - Launching DLSS4Test.exe with -sldebugoverlay causes SL to not function and the overlay does not appear
- New: 5426798 - [DLSS-RR] - Severe exposure corruption observed with different View Rect Offset options while DLSS-RR is enabled
- 5502683 - [Vulkan] - DLSS4Test-Win64-Shipping crashes on launch with non-NV GPUs (UE 5.6)
- 5502626 - [Vulkan] - DLSS4Test application exhibits heavy corruption on non-NV GPUs (UE 5.6)
- 5502713 - Application crash observed upon cycling through 'constrained view' and 'View Rect Offset' options with DLSS-SR performance modes (UE 5.6)
- 5417032 - [DX12] - Application launches to blank window on non-NV GPUs (UE 5.5)
- 5228046 - [RTX] - Image corruption occurs when View Rect Offset is adjusted
- 5502632 - Single view is the only functional Multi-Viewport option (UE 5.2)
- 5504701 - Verification HUD is showing "Enabled" for Reflex even after disabling it (UE 5.2)
- 5502698 - Corruption observed when changing "View Rect Offset" slider from 1 to 2 with RX enabled (UE 5.2)
- 5416464 - Application exhibits heavy corruption
- 5416489 - Application crashes upon changing DLSS-SR modes (UE 5.2)
- 5416646 - DLSS-SR is not engaged on launch despite 'auto' mode being selected
- 5416702 - [DX12][DLSS-FG] - Heavy stutter observed when changing constrained views with DLSS-FG enabled (UE 5.5)
- 5416990 - Only the DLSS4 Log populates when logging
- 5046562 - [DLSS-SR] - Reflection artifacts observed with DLSS-SR enabled in triple view
- 5046557 - Amplified instability observed on thin lines with different constrained views
- 5046507 - [DLSS-SR] - Amplified noise on reflections with DLSS-SR Enabled
- 5050485 - DLSS-SR + DLSS-RR UIs are functional with non-NV GPUs

8.3.0 September 2025

Component	Note
Unreal Engine	5.6 to 5.4 deprecated: 5.3, 5.2
NVIDIA RTX	5.6
UE Platforms	Win64, WinGDK
Streamline SDK	2.9.0
NGX SDK	310.4.0

RHI Support:

- D3D12RHI, D3D11RHI, VulkanRHI
    - DLSS Super Resolution (DLSS-SR)
    - Deep Learning Anti Aliasing (DLAA)
    - Image Scaling (NIS)
- D3D12RHI, D3D11RHI
    - RTX Dynamic Vibrance
    - Reflex Low Latency
- D3D12RHI
    - DLSS Ray Reconstruction (DLSS-RR)

    - DLSS Frame Generation (DLSS-FG)
    - DLSS Multi Frame Generation (DLSS-MFG)

Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

Note: The NVIDIA Image Scaling plugin is RHI agnostic, but might need adjustments for private platforms like consoles since only public RHIs are tested

Note: Support for deprecated UE versions will be removed in the future, but not before the next UE 5.X version release

New Features

- Updated to SL SDK 2.89.0
- Updated to NGX SDK 310.4.0
- Implement auto mode for modular DLSS upscaler

General Changes

- Deprecated support for UE 5.3, UE 5.2
- Added 'Getting Started with NVIDIA DLSS 4 for Unreal Engine.pdf' back, which was unintentionally missing from previous releases.
- Downgrade some Streamline SDK log messages to Verbose and VeryVerbose as they get routed to `LogStreamlineAPI`
- Use RDG_EVENT_SCOPE_STAT instead of deprecated RDG_GPU_STAT_SCOPE in version 5.6
- Added various functional tests
- Added *Incompatibilities with the Mobile Forward Renderer* sections to the plugin documentations

Breaking Changes

- Removal of DLSS-SR/RR sharpness
- Removed previously deprecated DLSS-SR and DLSS-RR presets
    - Projects which used those will be changed to use the respective default preset
- Fixed DLSS-SR not respecting alpha output by default. May cause a slight performance difference when alpha output is enabled. Set r.NGX.DLSS.EnableAlphaUpscaling 0 to restore old default behavior.

DLSS Super Resolution (DLSS-SR)

- Removed DLSS-SR presets A,B,C,D,E
- Removal of DLSS sharpness,
    - applications are encouraged to use NIS for sharpening, as outlined in the RTX UI Guidelines
    - Removed CVars
        - r.NGX.DLSS.Sharpness
        - r.NGX.DLSS.PreferNISSharpen
        - Removed Blueprint `UDLSSLibrary::SetDLSSSharpness` and `UDLSSLibrary::GetDLSSSharpness`
    - deprecated `UDLSSLibrary::GetDLSSModeInformation::OptimalSharpness` output so future integrations don't connect to it
        - The now deprecated `OptimalSharpness` always returns 0.35, which is what the underlying `NGXGetOptimalSettings` has been returning.
- Fixed DLSS-SR not respecting alpha output by default. May cause a slight performance difference when alpha output is enabled. Set r.NGX.DLSS.EnableAlphaUpscaling 0 to restore old default behavior.
    - CVar functionality change
    - r.NGX.DLSS.EnableAlphaUpscaling is now based off r.PostProcessing.PropagateAlpha by default (-1).
- Removed r.NGX.DLSS.AutomationTesting. The feature did not work as advertised. If you need to disable the NGX library from being loaded, use the -ngxdisable command line option or the r.NGX.Enable console variable.
- Added FDLSSModule::GetDLSSDenoiser(). Use this to work around other poorly behaved denoiser plugins that don't pass control down the denoiser chain when they're not active.

DLSS Ray Reconstruction (DLSS-RR)

- See DLSS-SR
- Removed DLSS-RR presets A,B,C
- On engine versions 5.4 and later, Lumen bilateral filtering is now automatically disabled while DLSS-RR is enabled in the blueprint library/methods.

DLSS Frame Generation (DLSS-FG)

Bug Fixes

- UE-1345 RenderGraphValidation Issue involving resource `CustomDepth` and the pass `BiasCurrentColorPass`

- 5416920 - [DLSS4 Test App] - 4k resolution option does not function correctly
- 5416658 - Editor crashes during second sequence of the Movie Render Queue (UE 5.2)

Known Issues

General:

- New: 5504512 - Heavily amplified instability observed with DLSS-SR Ultra Performance option (UE 5.6 & 5.5)
- 5417010 - Stand-alone game crashes on launch (UE 5.6)
- 5416670 - Stat DLSS average num continuously fluctuates with DLSS-SR enabled (5.5, 5.6)
- 5356855: Running the packaged sample in vulkan mode causes significant render instability or an immediate crash

Editor Specific:

- 5416522 - Unreal Engine Editor crashes when launched with Vulkan API (5.2)
- 5417003 - Editor crashes after MovieRenderQueue ends (5.6)
- 5417021 - Editor crashes after toggling through different DLSS-RR presets while playing the project in a new window (5.2)
- 5076953 Editor crashes with resolution/screen mode changes in Standalone Game mode with multi-display
- 4202194 - Unchecking "Warn about incompatible plugin tools" on VK does not remove the dialog box, even after restarting the editor

DLSS4Test App Specific:

- New: 5502683 - [Vulkan][DLSS4 Test App] - DLSS4Test-Win64-Shipping crashes on launch with non-NV GPUs (UE 5.6)
- New: 5502626 - [Vulkan][DLSS4 Test App] - DLSS4Test application exhibits heavy corruption on non-NV GPUs (UE 5.6)
- New: 5502713 - [DLSS4 Test App] - Application crash observed upon cycling through 'constrained view' and 'View Rect Offset' options with DLSS-SR - performance modes (UE 5.6)
- New: 5417032 - [DX12][DLSS4 Test App] - Application launches to blank window on non-NV GPUs (UE 5.5)
- New: 5228046 - [SDK/Plugin Release][DLSS4Test app][RTX] Image corruption occurs when View Rect Offset is adjusted
- New: 5416522 [Vulkan] - Unreal Engine Editor crashes when launched with Vulkan API (UE 5.2)
- New: 5502632 - [DLSS4Test-Win64-Shipping] Single view is the only functional Multi-Viewport option (UE 5.2)
- New: 5504701- [DLSS4Test App] Verification HUD is showing "Enabled" for Reflex even after disabling it (UE 5.2)
- New: 5502698 – [DLSS4Test App] Corruption observed when changing "View Rect Offset" slider from 1 to 2 with RX enabled (UE 5.2)
- 5416464 - Application exhibits heavy corruption
- 5416489 - Application crashes upon changing DLSS-SR modes (UE 5.2)
- 5416646 - DLSS-SR is not engaged on launch despite 'auto' mode being selected
- 5416702 - Heavy stutter observed when changing constrained views with DLSS-FG enabled (5.5)
- 5416990 - Only the DLSS4 Log populates when logging
- 5417032 - Application launches to blank window on non-NV GPUs (5.5, 5.6)
- 5228046 Image corruption occurs when View Rect Offset is adjusted
- 5046562 New Reflection artifacts observed with DLSS-SR enabled in triple view
- 5046557 Amplified instability observed on thin lines with different constrained views
- 5046507 Amplified noise on reflections with DLSS-SR Enabled
- 5050485 DLSS-SR + DLSS-RR UIs are functional with non-NV GPUs

8.2.0 July 2025

Component	Note
Unreal Engine	5.2 to 5.6
NVIDIA RTX	5.6

Component	Note
UE Platforms	Win64, WinGDK
Streamline SDK	2.8.0
NGX SDK	310.3.0

RHI Support:

- D3D12RHI, D3D11RHI, VulkanRHI
    - DLSS Super Resolution (DLSS-SR)
    - Deep Learning Anti Aliasing (DLAA)
    - Image Scaling (NIS)
- D3D12RHI, D3D11RHI
    - RTX Dynamic Vibrance
    - Reflex Low Latency
- D3D12RHI
    - DLSS Ray Reconstruction (DLSS-RR)
    - DLSS Frame Generation (DLSS-FG)
    - DLSS Multi Frame Generation (DLSS-MFG)

Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

Note: The NVIDIA Image Scaling plugin is RHI agnostic, but might need adjustments for private platforms like consoles since only public RHIs are tested

New Features

- Updated to SL SDK 2.8.0
- Updated to NGX SDK 310.3.0
- Support the IUpscalerModularFeature interface added in Unreal Engine 5.6 to allow other features to dynamically discover DLSS support when available.
    - Automatically adds DLSS Super Resolution/DLAA as an nDisplay plugin upscaling option with Unreal Engine 5.6 or later.

General Changes

- Deprecated DLSS-SR presets A,B,C,D,E
- Use the new slSetTag API with frame ids
    - Uses newer Streamline SDK API to tag resources for various frames in flight
    - Fallback to old behavior can be used with `Use slSetTag (deprecated)` plugin and plugin override setting as well as -sl{no}settag command line arguments
- Add plugin side support for upcoming NVRTX DLSS-RR enhancements
- StreamlineLogSink, wrap implementation in `#if !NO_LOGGING #endif` sandwich
- FStreamlineRHI::OnSwapchainCreated and FStreamlineRHI::OnSwapchainDestroyed, change Log verbosity to Verbose
- Add -sl{no}ota command line option to enable/disable OTA in addition to the project settings
    - Added plugin user ovveride setting
- Restructured and reorganized the DLSS-SR and DLSS-Project settings
- Updated DLSS-FG documentation about possible IQ issues with select engine features and plugins
- Updated DLSS-SR documentation about the difference between "engine needs run in TAA mode" vs the RTX UI guidelines suggesting to "disable temporal AA"

Breaking Changes

- `UDLSS::Auto` mode returned by `UDLSSLibary` BP library functions that haven't done so before. This means existing BPs that manually add UDLSS::Auto to UI elements might need ajustments to remove duplicate Auto modes in the UI

DLSS Super Resolution (DLSS-SR)

- Deprecated DLSS-SR presets A,B,C,D,E
- `UDLSSLibrary::GetSupportedDLSSModes` returns `UDLSSMode::Auto`
- `UDLSSLibrary::IsDLSSModeSupported(UDLSSMode::Auto)` returns `true`
- added note to *Enabling DLSS (Engine Side)* section about engine temporal AA mode vs RTX UI Guidelines "disable temporal AA mode"
- group DLSS-SR presets into separate DLSS-SR settings group

DLSS Ray Reconstruction (DLSS-RR)

- UE-1251 Add SSS and DoF guide buffers for DLSS-RR
- `UDLSSLibrary::GetSupportedDLSSModes` returns `UDLSSMode::Auto`
- `UDLSSLibrary::IsDLSSModeSupported(UDLSSMode::Auto)` returns `true`
- show paths of DLSS-RR binaries in the project settings to match those for DLSS-SR
- group DLSS-RR presets into separate DLSS-SR settings group

DLSS Frame Generation (DLSS-FG)

- Added *HDR and UI Color and Alpha* section to DLSS-FG documentation
- Added *Possible Reduced IQ with OpenColorUI plugin* section to DLSS-FG documentation

Bug Fixes

- UE-1222 Cvar r.Streamline.InitializePlugin declared too late.
- UE-1149 address clang compatibility and other issues reported by partners
- UE-1152 Fixes for const correctness and logging reported by partners
- 5158150 Enabling 'Load Debug Overlay' from the project settings causes a D3D device failure / GPU Crash with 'Standalone Game' mode

Known Issues

General:

- New: 5417010 - Stand-alone game crashes on launch (UE 5.6)
- New: 5416670 - Stat DLSS average num continuously fluctuates with DLSS-SR enabled (5.5, 5.6)
- 5356855: Running the packaged sample in vulkan mode causes significant render instability or an immediate crash

Editor Specific:

- New: 5416522 - Unreal Engine Editor crashes when launched with Vulkan API (5.2)
- New: 5417003 - Editor crashes after MovieRenderQueue ends (5.6)
- New: 5416658 - Editor crashes during second sequence of the Movie Render Queue (UE 5.2)
- New: 5417021 - Editor crashes after toggling through different DLSS-RR presets while playing the project in a new window (5.2)
- 5076953 Editor crashes with resolution/screen mode changes in Standalone Game mode with multi-display
- New: 4202194 - Unchecking "Warn about incompatible plugin tools" on VK does not remove the dialog box, even after restarting the editor

DLSS4Test App Specific:

- New: 5416464 - Application exhibits heavy corruption
- New: 5416489 - Application crashes upon changing DLSS-SR modes (UE 5.2)
- New: 5416646 - DLSS-SR is not engaged on launch despite 'auto' mode being selected
- New: 5416702 - Heavy stutter observed when changing constrained views with DLSS-FG enabled (5.5)
- New: 5416920 - 4k resolution option does not function correctly
- New: 5416990 - Only the DLSS4 Log populates when logging
- New: 5417032 - Application launches to blank window on non-NV GPUs (5.5, 5.6)
- 5228046 Image corruption occurs when View Rect Offset is adjusted
- 5046562 New Reflection artifacts observed with DLSS-SR enabled in triple view
- 5046557 Amplified instability observed on thin lines with different constrained views
- 5046507 Amplified noise on reflections with DLSS-SR Enabled
- 5050485 DLSS-SR + DLSS-RR UIs are functional with non-NV GPUs

8.1.0 July 2025

Note: New UE plugin version scheme to separate it from the DLSS 4 product version scheme.

Component	Note
Unreal Engine	5.2 to 5.6
NVIDIA RTX	5.6
UE Platforms	Win64, WinGDK
Streamline SDK	2.7.30

Component	Note
NGX SDK	310.2.0

RHI Support:

- D3D12RHI, D3D11RHI, VulkanRHI
    - DLSS Super Resolution (DLSS-SR)
    - Deep Learning Anti Aliasing (DLAA)
    - Image Scaling (NIS)
- D3D12RHI, D3D11RHI
    - RTX Dynamic Vibrance
    - Reflex Low Latency
- D3D12RHI
    - DLSS Ray Reconstruction (DLSS-RR)
    - DLSS Frame Generation (DLSS-FG)
    - DLSS Multi Frame Generation (DLSS-MFG)

Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

Note: The NVIDIA Image Scaling plugin is RHI agnostic, but might need adjustments for private platforms like consoles since only public RHIs are tested

New Features

- Support for UE 5.6
    - **Note**: DLSS plugin side changes to support nDisplay are not this release and will be added in the future
- Improved -d3ddebug and -rhivalidation compatibility for all supported UE versions
- Expose the "bias current color mask" DLSS-SR/RR inputs to allow UE applications to improve feature tracking

General Changes

- UE 5.6 support
    - UE 5.6 reworked the resource state handling between the D3D12RHI and RDG. The D3D12RHI resource state tracker got removed, and resource states are tracked by the RDG now. This required changes to the UE plugins to follow suit
    - UE 5.6 added new engine methods to allow the UE DLSS & Streamline plugin input and output resources to be made resident in memory. This avoids GPU access violations when the system is VRAM oversubscribed.
- Changes to UE 5.5 and older to improve to debug layer and memory residency compatibility, guided by the 5.6 changes
- Added *Graphics Debug and Validation Layers* sections to the Streamline and DLSS-SR/RR plugin documentation
- Consolidated various sections into new *Plugins and Platforms* section in the documentation
    - Added *Plugins and other Platforms* and *Plugins and Custom Engines* subsections
- Expose DLSS-SR/RR "bias current color mask" input to UE applications

DLSS Super Resolution (DLSS-SR)

DLSS may “erase” a particular feature or can produce a “ghost” or trail behind a moving feature. If a problematic asset is discovered during testing, one option is to instruct DLSS to bias the incoming color buffer over the colors from previous frames.

See *Enabling Bias Current Color Mask for DLSS* section in the DLSS-SR/RR UE plugin documentation for UE specifics.

DLSS Ray Reconstruction (DLSS-RR)

The DLSS-SR "bias current color mask" section also applies to DLSS-RR

Bug Fixes

- UE-1164 UE SL Reflex Plugin - Time spent in slReflexSleep is not treated as idle time, so the measured game thread timings will be incorrect when GPU limited with Reflex enabled.
- UE-1144 UE 5.5 Fix GPU memory access due to DLSS output texture not being resident
- 5048577 Application crash due to Error: Render subrect base and size exceed the dimensions of the input Color image
- 5228061 'Auto' Mode in the DLSS4Test app doesn't automatically adjust DLSS resolution, and only inherits the last set DLSS resolution

Known Issues

General:

- New: 5356855: Running the packaged sample in vulkan mode causes significant render instability or an immediate crash
- 5158150 Enabling 'Load Debug Overlay' from the project settings causes a D3D device failure / GPU Crash with 'Standalone Game' mode

Editor Specific:

- 5076953 Editor crashes with resolution/screen mode changes in Standalone Game mode with multi-display

DLSS4Test App Specific:

- 5228046 Image corruption occurs when View Rect Offset is adjusted
- 5046562 New Reflection artifacts observed with DLSS-SR enabled in triple view
- 5046557 Amplified instability observed on thin lines with different constrained views
- 5046507 Amplified noise on reflections with DLSS-SR Enabled
- 5050485 DLSS-SR + DLSS-RR UIs are functional with non-NV GPUs

4.0.2-2.7.30 April 2025

Component	Note
Unreal Engine	5.2 to 5.5
NVIDIA RTX	5.4
UE Platforms	Win64, WinGDK
Streamline SDK	2.7.30
NGX SDK	310.2.0

RHI Support:

- D3D12RHI, D3D11RHI, VulkanRHI
  - DLSS Super Resolution (DLSS-SR)
  - DLSS Ray Reconstruction (DLSS-RR)
  - Deep Learning Anti Aliasing (DLAA)
  - Image Scaling (NIS)
- D3D12RHI, D3D11RHI
  - RTX Dynamic Vibrance
  - Reflex Low Latency
- D3D12RHI
  - DLSS Frame Generation (DLSS-FG)
  - DLSS Multi Frame Generation (DLSS-MFG)

Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

New Features

DLSS Super Resolution

- Added a new default preset, Preset K (Transformer model)

DLSS Ray Reconstruction

- Added a new preset, Preset E (Transformer model)
- Deprecated DLSS-RR presets A,B,C

General Changes

Application developer facing:

- split the DLSS-RR presets off the DLSS-SR preset so they can be changed independently via cvars and project settings in the editor
- added support to detect deprecated DLSS-SR/RR presets and set those to default and then show an editor message box & log message.
- add tooltips to the DLSS-SR/RR presets based on the descriptions in the SDK docs & headers

Plugin internal:

- don't create empty NGX log folders
- removed r.NGX.EnableOtherLoggingSinks

DLSS Frame Generation (DLSS-FG)

- added cvar for eRetainResorcesIfOff flag that can be used to reduce stutter when changing DLSS-FG settings in a UI

DLSS Super Resolution (DLSS-SR)

A new Preset K project setting has been added to select the DLSS-SR Transformer model

DLSS Ray Reconstruction (DLSS-RR)

DLSS Ray Reconstruction supports stock UE to a certain degree. However the engine modifications in the NVRTX 5.4 branch additional input buffers which will greatly improve the overall quality of DLSS-RR.

A new Preset E project setting has been added to select the DLSS-RR Transformer model

Presets A,B,C have been deprecated and been hidden from the project settings. They can still be selected via the `r.NGX.DLSSRR.Preset` cvar but it is not recommended to do so.

The experimental Substrate material system is not compatible with DLSS-RR. DLSS-RR uses the traditional G-Buffer which is used to compute guide buffers such as diffuse and specular albedo. Substrate however does not provide a traditional guide Buffer. This will be addressed in a future release.

Bug Fixes

- UE-1111 DLSS-SR Preset J cannot be selected due to type casting but
- UE-1119 Crash in StreamlineCore when the deprecated Streamline plugin is not present
- UE-1122 In shutdown NGX check for StreamlineCore instead of Streamline plugin
- UE-1124 StreamlineLatewarp: fix 'bDrawInGame': is not a member of 'FPrimitiveSceneInfo'
- UE-1113 Ensure StreamlineReflex detects its own tick rate handler
- UE-1132 UStreamlineLibraryReflex::GetReflexMode() return Invalid enum when mode is EStreamlineReflexMode::Boost
- UE-1150 FTexture2DRHIRef deprecation warnings for the NIS plugin
- UE-1110 Fix FTexture2DRHIRef deprecation warnings StreamlineLatewarp

Known Issues

General:

- New: 5158150 Enabling 'Load Debug Overlay' from the project settings causes a D3D device failure / GPU Crash with 'Standalone Game' mode
- New: 5048577 Application crash due to Error: Render subrect base and size exceed the dimensions of the input Color image

Editor Specific:

- 5076953 Editor crashes with resolution/screen mode changes in Standalone Game mode with multi-display

DLSS4Test App Specific:

- New: 5228061 'Auto' Mode in the DLSS4Test app doesn't automatically adjust DLSS resolution, and only inherits the last set DLSS resolution
- New: 5228046 Image corruption occurs when View Rect Offset is adjusted
- 5046562 New Reflection artifacts observed with DLSS-SR enabled in triple view
- 5046557 Amplified instability observed on thin lines with different constrained views
- 5046507 Amplified noise on reflections with DLSS-SR Enabled
- 5050485 DLSS-SR + DLSS-RR UIs are functional with non-NV GPUs

4.0.0-2.7.3 February 2025

Component	Note
Unreal Engine	5.2 to 5.5

Component	Note
NVIDIA RTX	5.4
UE Platforms	Win64, WinGDK
Streamline SDK	2.7.3
NGX SDK	310.1.0

RHI Support:

- D3D12RHI, D3D11RHI, VulkanRHI
  - DLSS Super Resolution (DLSS-SR)
  - DLSS Ray Reconstruction (DLSS-RR)
  - Deep Learning Anti Aliasing (DLAA)
  - Image Scaling (NIS)
- D3D12RHI, D3D11RHI
  - RTX Dynamic Vibrance
  - Reflex Low Latency
- D3D12RHI
  - DLSS Frame Generation (DLSS-FG)
  - DLSS Multi Frame Generation (DLSS-MFG)

Note: The VulkanRHI is not implemented in general for the Streamline family of plugins, only for the NGX based DLSS plugins

New Features

- DLSS Frame Generation
  - Added support for Multi-Frame Generation
- DLSS Super Resolution
  - Added a new preset, Preset J (Transformer model)
- DLSS Ray Reconstruction
  - General availability
  - Added a new preset, Preset D (Transformer model)

General Changes

Application developer facing:

- The architecture of the family of Streamline plugins has been changed in the 4.0.0 release compared to the 1.x.x releases:
  - Plugins have been renamed
  - Implementation has been split into separate plugins.
  - Some blueprint enums and class names have been changed for UE best practices.
- Throw Blueprint Exception when an invalid BP enum value is passed into BP functions (5.4+)
- Added WinGDK support

Plugin internal:

- Add r.Streamline.FilterRedundantSetOptionsCalls and -slfilter -slnofilter command line options
- NGX DLSS plugin interop at slInit time: only pass in paths to the DLSS-SR/RR NGX binaries when r .NGX.Enable/r.NGX.Enable.AllowCommandLine etc would not prohibit doing so on the NGX DLSS plugin.
- Add r.Streamline.LogFunctions.LogLevel to customize which functions are logged when r.Streamline.LogFunctions is 1
- Log as the UE Error category when Streamline interpose cannot be loaded due to missing .dll or similar
- Add r.Streamline.Load.* console variables to specific which underlying Streamline binaries are loaded at slInit time
- Add support for specular motion vectors (NVRTX 5.4+)

Bug Fixes

- Fix bug of DLSS-FG enganging unintendedly on notification windows when PIE (This Viewport) by correctly detecing PIE (New Window) where DLSS-FG is supported

Known Issues

Editor Specific:

- 5076953 Editor crashes with resolution/screen mode changes in Standalone Game mode with multi-display

DLSS4Test App Specific:

- 5048577 Application crash observed upon toggling different rect offset options
- 5046562 New Reflection artifacts observed with DLSS-SR enabled in triple view
- 5046557 Amplified instability observed on thin lines with different constrained views
- 5046507 Amplified noise on reflections with DLSS-SR Enabled
- 5050485 DLSS-SR + DLSS-RR UIs are functional with non-NV GPUs

Upgrading from a project from 1.x.x plugins to 4.x.x plugins

The architecture of the family of Streamline plugins has been changed in the 4.0.0 release compared to the 1.x.x releases:

- Plugins have been renamed
- Implementation has been split into separate plugins
- Some of the blueprint enums and class names have been changed between 1.x.x and 4.x.x to follow overall UE best practises.

Blueprint redirectors have been added to the 4.0.0 release to reduce the friction of upgrading. If those fail for whatever reason the mapping should be straightforward.

Previous 1.x.x plugin architecture

In the *previous* release the 1.x.x. Streamline plugin contained the core of the implementation that connects the engine with the underlying Streamline APIs.

On the developer facing Blueprint side there were a set of Blueprint libraries for generic (`UStreamlineLibrary`) as well as set of feature specific use(such as `UStreamlineLibraryDLSSG` or `UStreamlineLibraryReflex`).

Current 4.x.x plugin architecture

In the *current* 4.x.x releases, the core implementation parts from the 1.x.x Streamline plugin has been renamed to the 4.x.x StreamlineCore plugin. The generic Blueprint library `UStreamlineLibrary` continues to be part of the renamed StreamlineCore plugin.

Note: StreamlineCore is hidden in the editor intentionally since the per feature plugins implicitly enable it.

The feature specific Blueprint libraries from the 1.x.x Streamline plugin however have moved to a set of 4.x.x feature specific plugins such as StreamlineDLSSG or StreamlineReflex.

Note: StreamlineDeepDVC Blueprint libraries were already separated out in 1.x.x into a separate StreamlineDeepDVC plugin, but now the other features are also separated.

Changes to Blueprint Enum Values

As part of DLSS-MFG support, the enum values of the various enums have changed. This should not impact any well behaved client blueprints that treat enums as opaque identifiers. But any blueprint or C++ code that casts a byte/integer into enum, and thus side-stepping the type safety of Blueprints & C++, will get this reported as a scripting error: `LogScript: Error: Script Msg: UStreamlineLibraryDLSSG::SetDLSSGMode should not be called with an invalid enum value (252) "EStreamline DLSSGMode MAX"`

The fix is to populate the UI element with a list of the enums returned from `UStreamlineLibraryDLSSG::GetSupportedDLSSGModes` and then index into that array via the UI element selection index to get the proper EStreamlineDLSSGMode argument for the call to `UStreamlineLibraryDLSSG::SetDLSSGMode`

Note: `EStreamlineReflexMode` always has had non-contiguous enum *values*, so similar issues might have arosen in the past. A `UStreamlineLibraryReflex::GetSupportedReflexModes` was added in the 4.0.0 release to allow for a similar pattern as for DLSS-FG when populating UE elements with the supported Reflex Low Latency Modes.

Upgrading a 1.x.x project to 4.x.x plugins

The NVIDIA Streamline 1.3.3 Legacy Compatibility plugin (Streamline.uproject) doesn't contain any code and only enables the following plugins to allow projects developed against the 1.x.x family of plugins to continue to work.

- StreamlineCore
- StramlineDeepDVC
- StreamlineReflex
- StreamlineDLSSG

For the 4.x.x family of plugins, the recommended alternative is to explicitely enable the plugins for each required feature.

Note: Streamline.uplugin is hidden in the editor intentionally since it exists purely for existing projects and is not required for new projects

DLSS Generation (DLSS-FG, DLSS-MFG)

The DLSS Multiframe Generation modes have been added as `3X` and `4X` members to EStreamlineDLSSGMode, `2X` corresponds to the On mode of the 1.x.x release.

Any existing BP or C++ code that has been using `GetSupportedDLSSGModes` and `UStreamlineLibraryDLSSG::SetDLSSGMode` to implement UI should require no additional effort (other than maybe localization) to have DLSS-MFG be available in a project.

DLSS Super Resolution (DLSS-SR)

A new Preset J project setting has been added to select the DLSS-SR Transformer model

DLSS Ray Reconstruction (DLSS-RR)

DLSS Ray Reconstruction supports stock UE to a certain degree. However the engine modifications in the NVRTX 5.4 branch additional input buffers which will greatly improve the overall quality of DLSS-RR.

A new Preset D project setting has been added to select the DLSS-RR Transformer model

`UDLSSLibrary::IsRayTracingAvailable()` has been added to make implementing UI for DLSS-RR easier

Reflex Low Latency

`UStreamlineLibraryReflex::GetSupportedReflexModes()` has been added to match other BP libraries and make generation UIs easier

1.3.3-SL2.4.15

Release Date: September 2024

- Bug fixes
  - replace viewport tracking related asserts with log messages
  - gracefully handle VR/XR by not asserting
  - actually limit the number of generated swapchain proxies depending on the cvar
- Other Changes
  - Update to Streamline 2.4.15
  - removed UE4 support in the editor (PIE) due to instability reated to thread (un)safe accessing of Slate TSharedPtrs
  - enable multi viewport support for DLSS-FG
  - partial 5.5 compatibility
  - various refactorings in the BP libraries to reduce duplicate code
  - add new modes to r.Streamline.DLSSG.AdjustMotionBlurTimeScale to handle auto mode

1.3.2-SL2.4.10

Release Date: June 2024

- Other changes
  - Update to Streamline 2.4.10
  - Support targetting minimum Windows version > Windows 7 (WinGDK platform support)
  - Disable the UE 5.4 workaround when not needed

1.3.1-SL2.4.0

Release Date: May 2024

(Not released on the market place)

- Bug fixes
  - Fix crash on editor startup with D3D11
  - Fix parsing of -slswapchainprovider command line option
  - Fix incorrect public dependencies in StreamlineBlueprint module
- Other changes
  - Support UE 5.4 release

1.3.0-SL2.4.0

Release Date: April 2024

- New features
  - RTX Dynamic Vibrance (DeepDVC)
  - DLSS-FG subrect support
  - Option to enable DLSS-FG dynamic resolution
  - Option to automatically disable DLSS-FG if full screen menus are detected
- Bug fixes
  - Fix assertion with MovieRenderQueue
  - Fix assertion when taking screenshots with F9
  - Fix hudless texture extent
  - Fix viewport resize issue
  - Fix assert when running dumpgpu command
  - Only compile shaders on supported platforms, fixes some console builds
- Other changes
  - Update to Streamline 2.4.0
  - Split PCL and Reflex
  - Removed support for SetInputSampleLatencyMarker
  - SL function logging improvements

1.2.0-SL2.2.1

Release Date: October 2023

- New features

    - DLSS-FG auto mode
- Bug fixes
    - Fix startup crash when -RenderOffScreen is used
    - Support running from network paths beginning with \\
- Other changes
    - Update to Streamline 2.2.1
    - Log plugin version on startup
    - Improve error log on DXGI/D3D12 errors from Streamline

1.1.0-SL2.2.0

Release Date: September 2023

- New features
    - Add r.Streamline.InitializePlugin read-only cvar to allow disabling plugin
    - Add FStreamlineCoreModule::QueryDLSSGSupport()
- Bug fixes
    - Fix assertion when DXGI_STATUS results are reported by SL SDK. Could happen when app loses focus
- Other changes
    - Support UE 5.3.0 release
    - Disable DLSS-FG when window is not in focus

1.0.7-SL2.2.0

Release date: August 2023

- New features
    - none
- Bug fixes
    - none
- Other changes
    - Update to Streamline 2.2.0
    - Source-level support for UE 5.3.0 preview 1

1.0.6-SL2.1.1

Release date: July 2023

- New features
    - Add r.Streamline.TagUIColorAlphaThreshold cvar for UI extraction pass alpha threshold value
    - Add new blueprint libraries to express various end user recoverable ways why DLSS-FG failed to the game logic
    - Add min sup ported viewport width and an inaccurate vram estimate to "stat dlssg"
    - Add error handler for DLSS-FG runtime errors
    - Add r.Streamline.DLSSG.CheckStatusPerFrame to check the DLSS-FG status at runtime and assert if it's failing somehow (default = true)
- Bug fixes
    - Fix crash seen during shutdown when SWindow doesn't have a Viewport
    - Disable DLSS-FG if the viewport width is below what DLSS-FG supports at the SL level
    - Fix minor leak in reflex hooks
- Other changes
    - Update to Streamline 2.1.1
    - Disable DLSS-FG on scene captures

1.0.5

- there was no version 1.0.5 plugin release

1.0.4-SL2.1.0

Release date: June 2023

Initial public plugin release